



CAP182_SS2_PART D



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1. Model Selection

Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	89.34%	65.56%	18.71%	29.12%	77.89%
Gradient Boosting	89.52%	67.08%	20.42%	31.30%	79.79%

Recommendation

Based on the results, Gradient Boosting (Model 2) is recommended as the preferred model for further development.

Gradient Boosting achieved higher scores than Logistic Regression across all evaluation metrics, including accuracy, precision, recall, F1-score and ROC-AUC. Although the differences are relatively small, the improvement was consistent across all measures, indicating stronger overall predictive performance. Particular attention should be given to the ROC-AUC score, which measures how well the model distinguishes between customers who convert and those who do not. Gradient Boosting achieved a ROC-AUC of 79.79%, compared with 77.89% for Logistic Regression, suggesting a better ability to identify customers who are likely to convert.

As the dataset is imbalanced, accuracy should be interpreted with caution. Approximately 88% of observations belong to the non-conversion class, while only 12% belong to the conversion class. In this context, precision and recall provide more meaningful insights into model performance. Gradient Boosting achieved a precision of 67.08% and a recall of 20.42%, outperforming Logistic Regression on both measures. While the improvement in recall is modest, it is particularly relevant to STADIOEquities because recall reflects the proportion of customers who eventually convert that could potentially be identified for targeted intervention. However, the relatively low recall suggests that many potential converters are still being missed, indicating that further model refinement would be beneficial.

Gradient Boosting is therefore recommended because it provides the strongest overall predictive performance while maintaining reasonable precision. However, the model should be viewed as a development recommendation rather than a final production solution, as the results are based on the UCI Bank Marketing proxy dataset and should be re-evaluated using actual STADIOEquities data.

2. Model Improvement

Although Gradient Boosting achieved the strongest overall performance, the results indicate considerable room for improvement. The most significant limitation is the relatively low recall of 20.42%, meaning that many positive cases are still not being identified. The following improvements could be implemented to improve recall and enhance the model's overall effectiveness:

a) Hyperparameter Tuning

The current model was developed using largely default settings. Parameters such as the number of boosting stages, learning rate and tree depth can influence model performance. A structured search process, such as Grid Search combined with cross-validation, could be used to evaluate different parameter combinations and identify a configuration that improves predictive performance while maintaining the model's ability to generalise to unseen data.

b) Classification Threshold Optimisation

The current model uses the default threshold of 0.50. Lowering the threshold may increase recall by identifying more positive cases, although this would likely reduce precision. The optimal threshold should therefore be determined using business considerations, such as the cost of missed activation opportunities and the resources available for targeted customer interventions.

c) Additional Feature Engineering

The proxy dataset used in this study does not contain the same information that would be available within the STADIOEquities environment. The requested data includes customer characteristics, onboarding progress, app and web behaviour, deposit attempts, marketing engagement and customer support information. These variables are likely to provide more direct indicators of first deposit activation and may improve the model's ability to identify customers who are at risk of not activating.

Finally, any revised model should be evaluated using the same stratified cross validation and held-out test approach used in this study to ensure that improvements in performance reflect genuine predictive value rather than overfitting to the training data.

3. Adaptation to STADIOEquities Data

The selected Gradient Boosting model would need to be adapted to the structure and definitions of the actual STADIOEquities data before it could be used to predict first-deposit activation. The six requested datasets would first need to be integrated using a consistent anonymised Customer ID, preferably in UUID format, with all date and time fields recorded consistently in UTC. Categorical variables such as Risk Appetite, Deposit Method, Acquisition Channel, Issue Type and Resolution Status should also have clearly defined and consistent valid values across the datasets.

The model would then need to be retrained using the actual STADIOEquities data rather than the Bank Marketing proxy dataset. This would involve creating the target outcome based on whether a customer successfully makes a first deposit and using information from the customer, onboarding, behavioural, funding, marketing and support datasets as potential predictors of activation. The requested STADIOEquities data is likely to provide more direct indicators of activation behaviour than those available in the proxy dataset. Variables such as onboarding progress, deposit attempts, deposit methods, marketing engagement and customer-support interactions may provide valuable insight into why some customers activate while others remain unfunded.

Once the data has been prepared, the Gradient Boosting model should be retrained and evaluated using the same cross validation and held-out test approach applied in this study. This will allow STADIOEquities to determine whether the relationships identified in the proxy dataset remain relevant when applied to its own customer base and activation process.

4. Alignment with Related Literature

The results of this study are broadly consistent with the related literature. Verster et al. (2021) found that tree-based ensemble models outperformed Logistic Regression when predicting home-loan take-up, while Gradient Boosting achieved stronger performance than Logistic Regression across all evaluation metrics in this study. Similarly, Breed and Verster (2017) reported that Gradient Boosting achieved the highest validation Gini coefficient on a South African banking direct-marketing dataset, providing further evidence that boosting techniques can be effective for modelling customer conversion behaviour. The results are also consistent with Moro, Cortez and Rita (2014), whose study used the same Bank Marketing dataset selected as the proxy dataset for this analysis. Their study compared multiple classification techniques and applied feature selection before evaluating model performance. Similar principles were applied in this study through model comparison and evaluation using cross-validation and a held-out test dataset.

However, some differences should be noted. The studies reviewed in Part A investigated home loan take-up, direct-marketing response and term deposit subscriptions, whereas this project focuses on first deposit activation. The datasets, variables and evaluation measures also differed, with the literature primarily reporting Gini coefficients, while this study focused on Accuracy, Precision, Recall, F1-score and ROC-AUC.

Breed and Verster (2017) further concluded that no single modelling technique consistently performs best across all datasets. This is an important consideration for STADIOEquities. Although Gradient Boosting performed best in this study and is consistent with findings from some of the related literature, the model should still be retrained and validated using actual STADIOEquities customer data before final conclusions about first-deposit activation are drawn.

References

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